

Fine-Grained Zero-Shot Vehicle Make and Model Recognition: A Survey and Future Directions

Adriana-Victoria Miu (Pavel)
National University of Science and Technology
POLITEHNICA Bucharest
Bucharest, Romania
adrianamivictoria55@gmail.com

Andrei Vizureanu
National University of Science and Technology
POLITEHNICA Bucharest
Bucharest, Romania
andrei.vizureanu@gmail.com

Abstract—Vehicle Make and Model Recognition (VMMR) is a fine-grained visual classification challenge situated in demanding real-world conditions—low resolution, occlusion, adverse illumination, and arbitrary viewpoints typical of traffic surveillance footage. Current state-of-the-art systems rely on closed-set supervised classifiers that fail when confronted with unseen makes, model-year facelifts, or newly introduced brands, yet Intelligent Transportation Systems (ITS) require recognition capability that scales seamlessly as automotive lineups evolve. This paper presents a structured literature review tracing VMMR from hand-crafted feature pipelines through part-aware convolutional and vision transformer architectures to contrastive vision-language models and systematically maps this trajectory against Zero-Shot Learning (ZSL) and Generalized ZSL (GZSL) developments. Our synthesis produces: (i) a taxonomy of VMMR method families organized by supervision regime and open-world readiness; (ii) a characterization of evaluation protocols spanning closed-set top-1 accuracy to GZSL harmonic mean across web-sourced and surveillance-domain dataset splits; and (iii) identification of recurring design patterns—geometric normalization, hierarchical coarse-to-fine scoring, part-aware attention, and semantic prompt alignment—that collectively define the path toward open-vocabulary vehicle recognition. Our cross-analysis reveals that contrastive vision-language models such as SigLIP2 and CLIP represent the most promising foundation for zero-shot VMMR, provided five documented gaps are addressed: standardized GZSL benchmarks for vehicle datasets, domain-specific automotive prompt engineering, hierarchical make-aware inference, surveillance-domain robustness, and edge-deployment efficiency.

Index Terms—vehicle make and model recognition; zero-shot learning; open-set recognition; fine-grained visual recognition; vision transformers; contrastive learning; intelligent transportation systems; object recognition

I. INTRODUCTION

Vehicle Make and Model Recognition (VMMR) is an established problem in Intelligent Transportation Systems (ITS) with applications spanning traffic analytics, congestion management [1], law enforcement and stolen-vehicle detection [2], and situational awareness in autonomous driving. Its core motivation is complementarity with license-plate recognition: plates can be occluded, altered, or spoofed, whereas the structural appearance of a vehicle—grille topology, headlamp cluster, hood profile, and badge placement—cannot be readily changed. Identifying make and model further unlocks secondary vehicle attributes (production year, dimensions, powertrain) that enrich ITS data fusion pipelines [1].

From a computer-vision perspective, VMMR is a fine-grained intra-category classification problem complicated by three compounding factors: (i) high inter-class similarity, as different makes converge on aerodynamic forms and shared platform architectures; (ii) substantial intra-class variance

across trim levels, model-year facelifts, and colors; and (iii) operational capture degradation from arbitrary viewpoints, low resolution, adverse weather, motion blur, and partial occlusion [3]. Front-view imagery is often the most discriminative and is commonly used in VMMR, as the frontal zone concentrates the most differentiating visual evidence; some works also leverage rear or multi-view inputs where camera geometry permits.

Despite continuous benchmark improvements, all current VMMR systems share a structural limitation: closed-set operation, requiring that every test-time class appeared during training. The automotive market continuously introduces new vehicles—new regional entrants, annual refreshes, and rapidly proliferating electric vehicle brands—making perpetual re-annotation unsustainable. ZSL addresses this directly: by aligning visual features with semantic side information such as class attributes, textual descriptions, or brand ontologies, a ZSL system can recognize classes absent from training images [4], [5]. This survey provides a systematic cross-examination of VMMR and ZSL, mapping their intersection and deriving a concrete research agenda for operational zero-shot VMMR.

II. VMMR METHOD FAMILIES AND TRENDS

A. Traditional Feature Pipelines

Pre-deep VMMR relied on hand-crafted local descriptors with bag-of-words encoding and linear classifiers. Jamil et al. [6] proposed Bag-of-Expressions (BoE), extending standard Bag-of-Words by encoding spatial co-occurrence geometry (Fig. 1) among visual words derived from KAZE, SURF, FAST, and BRISK keypoints described with Histogram of Oriented Gradients (HOG). Expression dictionaries capture pairwise neighbor relationships, yielding greater robustness to viewpoint change and occlusion than isolated descriptors; combined with multiclass linear Support Vector Machines (SVMs), BoE achieved state-of-the-art accuracy among pre-deep baselines on NTOU-MMR while maintaining CPU-level throughput [6].

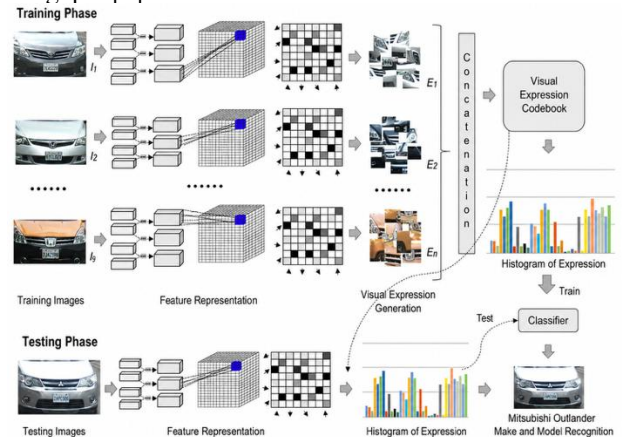


Fig. 1. Bag of expressions-based framework for vehicle make and model recognition (VMMR) [6]

Siddiqui et al. [7] complemented this with a compact HOG-plus-GIST pipeline (1,152-D HOG, 512-D GIST over four scales), reaching 97.89% accuracy at 13.9 fps via one-vs-one SVM on a standard CPU—a practical real-time baseline for settings where plate evidence is unavailable.

Takeaway: Traditional pipelines demonstrate the feasibility of CPU-deployable VMMR with interpretable features. Their fundamental limitation is a fixed vocabulary: the classifier cannot produce a label for any class outside its training taxonomy, making them entirely closed-set. The performance ceiling under long-tail class distributions and significant domain shift (lab images vs. surveillance footage) further motivates deeper architectures.

B. CNN-Based Fine-Grained Recognition

Deep convolutional neural networks (CNNs) transformed VMMR accuracy. Sochor et al. [8] (BoxCars) exploited fixed surveillance camera geometry to auto-calibrate 3D bounding boxes, unpacking vehicle surfaces into planar patches and injecting viewpoint vectors as auxiliary CNN channels. On BoxCars (21,250 vehicles, 148 fine-grained classes), this geometry-aware strategy reduced classification error by 26% and lifted verification average precision by 208%, establishing that explicit pose normalization is critical under diverse surveillance viewpoints [8]. Lyu et al. [9] introduced a Two-Branch Two-Stage (2B-2S) (Fig. 2) architecture separating make-level (coarse) and model-level (fine) classification, with a decision module quantifying inter-make confusion reduction via a Gain score. On DVMM (23 makes, 326 European-market models), 2B-2S reached 93.95%, consistently outperforming single-branch baselines [9]. Lu et al. [10] recovered part-level discriminative cues without manual annotations using dilated convolutions and cross-channel grouping, achieving 94.5% on Stanford Cars at approximately 140 fps. Amirkhani and Barshooi [3] formalized part-based multi-agent inference in DeepCar 5.0 (Fig. 3): four CNN agents for headlights, upper grille, bumper, and fog regions vote by blackboard consensus, reaching 92.14% on 40,185 images under occlusion and adverse weather. Nafzi et al. [11] tackled long-tail scale challenge by harvesting multi-year, multi-view images, inferring year-view subcategories through hierarchical clustering, expanding the taxonomy to 4,730 classes, and lifting surveillance top-1 accuracy from 29.0% to 71.1%.

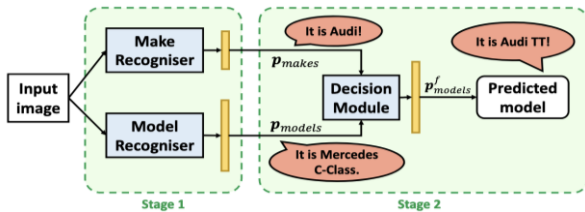


Fig. 2. The scheme of the proposed Two-Branch–Two-Stage (2B–2S) framework for VMMR [9]

Takeaway: CNN-based methods demonstrate that geometric normalization, part-aware attention, and coarse-to-fine decomposition each address distinct subproblems within VMMR. Despite these advances, all systems remain closed-set—cross-entropy training with softmax over a fixed class set produces embeddings discriminative within the training taxonomy but poorly calibrated for unseen inputs.

They also demand expensive retraining cycles whenever new makes or models are introduced, motivating semantic alignment approaches.

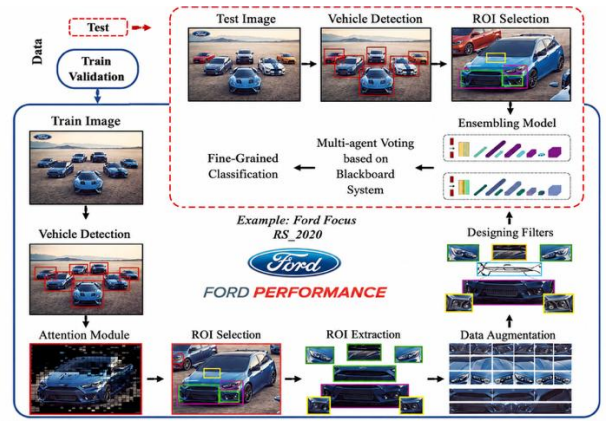


Fig. 3. Multi-agent VMMR flowchart proposed in [3]

C. Transformer and Vision-Language Approaches

Vision Transformers (ViT) [12], introduced by Dosovitskiy et al., process images as patch sequences under global self-attention, capturing long-range dependencies that local convolutional filters miss. Taki and Zemmouri [13] showed a fine-tuned ViT-Base model achieves 99.3% on a vehicle type classification benchmark, surpassing VGG16 (99.2%) and CNNs trained from scratch (92%), attributable to pretraining scale and global attention. Liu et al. [14] extended the finding to carrier-free UWB radar echoes converted to 2D images: ViT outperforms LeNet, VGG16, and ResNet50 across signal-to-noise levels, demonstrating that transformer attention captures cross-azimuth patterns more robustly. DINOv2 [15] further strengthens this direction by providing dense patch-level features trained self-supervisedly without labels, yielding strong zero-shot retrieval across domains.

The first direct evaluation of a vision-language model (VLM) on VMMR was conducted by Kerdvibulvech [16], applying GPT-4V to CompCars in a purely zero-shot setting (exposed in Fig. 4). Results confirm that large multimodal models carry latent brand-identification capability but degrade significantly with multiple vehicles per image and exhibit systematic errors on Chinese and regionally under-represented brands—a documented consequence of training corpus geographic bias. Many out-of-distribution inputs were assigned UNKNOWN or misattributed to visually similar makes. This establishes a concrete empirical baseline: VLMs possess the semantic grounding required for VMMR, but targeted adaptation is necessary [16].

Takeaway: Transformer backbones dramatically improve feature quality and global context modeling. VLMs take this further by connecting visual features to natural language, enabling open-vocabulary queries. The GPT-4V experiment confirms this potential while exposing brand bias and multi-object failures as the primary barriers—precisely the gaps that structured ZSL methods are designed to close.

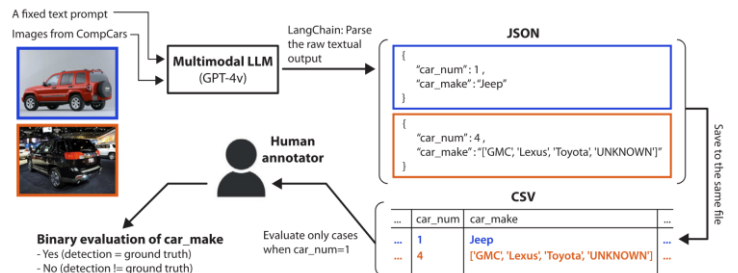


Fig. 4. The workflow of [16] with example results from GPT-4v

D. Datasets and Evaluation Protocols

Benchmark datasets fall into two families. Web-sourced datasets—Stanford Cars (196 classes), CompCars (web-nature; 1,716 models), and VMMDb (495 make-model combinations)—provide high-resolution, clean images covering diverse models, but their capture conditions differ substantially from real deployments. Surveillance-oriented datasets—BoxCars (21,250 vehicles; 148 fine-grained classes from a fixed outdoor camera), NTOU-MMR, DVMM (326 European models), and DeepCar 5.0 (40,185 frontal images; 480 classes; 50 automakers)—target operational realism with occlusion, variable lighting, and lower resolution [8], [9], [3].

Evaluation protocols are equally diverse and often incomparable across papers. Closed-set top-1 and top-5 accuracy measure discrimination within the training taxonomy but provide limited insight into generalization.

Open-set protocols such as the BoxCars verification task [8] assess pairwise similarity and rejection of unknown instances, approximating (but not fully implementing) open-world recognition. For zero-shot and generalized zero-shot evaluation, the field reports per-class accuracy on unseen classes (conventional ZSL) and the harmonic mean of

seen/unseen accuracies under joint evaluation at inference time (GZSL) [4], [5]. Critically, no VMMR benchmark currently provides standardized ZSL splits with semantic class descriptors, limiting progress toward comparable open-world evaluation.

Table I summarizes representative VMMR-related datasets, emphasizing scale, imaging conditions, and practical readiness for open-world evaluation. In addition to make-model datasets, we include large-scale vehicle re-identification (ID-based) datasets because they provide abundant surveillance-domain imagery and multi-frame/multi-camera observations that can support representation learning, domain-robust pretraining, and track-level aggregation—capabilities transferable to VMMR and zero-shot VMMR.

Takeaway: Public VMMR datasets rarely provide standardized seen/unseen partitions or semantic descriptors, and web-to-surveillance domain shift complicates cross-paper comparison—supporting the need for a unified GZSL protocol (Gap G1).

TABLE I. Representative vehicle datasets (VMMR and re-identification) and their suitability for open-world/ZSL evaluation.

Dataset	Classes	Images	Resolution range	Viewpoints	Capture conditions	Public availability	ZSL splits exist?
Stanford Cars [17]	196	16,185	High (web)	Multi-view	Web	Yes	No (adaptable)
CompCars (web-nature) [18]	163 makes / 1,716 models	136,726 full-car + 27,618 parts	Med-high	Multi-view (labeled)	Web	Research license (request)	No (adaptable)
CompCars (surveillance-nature) [18]	Model labels	50,000	Low-med	Front-view (fixed)	Surveillance	Research license (request)	No
BoxCars116k [8]	693	116,286	Low-med	Fixed roadside	Surveillance	Research/limited	No
VehicleID [19]	ID-based	221,763	Low-med	Rear/mixed	Surveillance	Yes	No (ID task)
VeRi-776 [20]	ID-based	49,357	Low-med	Multi-camera	Surveillance	Yes	No (ID task)
VERI-Wild [21]	ID-based	416,314	Low	Multi-camera	Surveillance	Yes	No (ID task)
CityFlow [22]	ID-based	229,680	Low-med	Multi-camera	Surveillance	Yes	No (ID task)
VMMDb [23]	9,170	291,752	Med-high	Multi-view	Web	Research/limited	No
DeepCar 5.0. [3]	480	40,185	Med	Frontal bias / mixed	Mixed (real traffic)	Research/limited	No
NTOU-MMR [24]	29	5,835	Low-med	Multi-view	Web / mixed	Yes	No
DVMM [9]	326	228,463	Med-high	Multi-view	Web	Research/limited	No

III. ZERO-SHOT LEARNING: METHODS AND TAXONOMY

A. Foundations and GZSL Formulation

ZSL was formalized around aligning visual features to semantic class prototypes—initially attribute vectors or word embeddings. Zhang and Saligrama [25] proposed Semantic Similarity Embedding (SSE), representing each unseen class as a convex combination of seen-class embeddings, enabling recognition through structural similarity transfer. Their subsequent deep embedding framework [26] paired convolutional features with compatibility functions trained by max-margin objectives. Two persistent failure modes emerged: hubness, where a few semantic prototypes dominate nearest-neighbor retrieval in high dimensions; and seen-class bias in the GZSL setting, where the model over-predicts training classes when seen and unseen classes are evaluated jointly at inference time [4]. Rahman et al. [5] provided a unified framework bridging ZSL, GZSL, and few-shot learning (FSL) through class-adapting projection matrices. Standard GZSL metrics are per-class top-1 accuracy for

conventional ZSL and harmonic mean H for GZSL, where both seen and unseen accuracy must be balanced [4], [5].

B. Semantic Refinement Strategies

To address prototype drift and hubness, targeted refinement strategies were developed. Kim et al. [27] proposed explicit extraction of attribute-relevant features while suppressing irrelevant visual information, improving transferability on fine-grained benchmarks including CUB and SUN. Zhang and Zhang [28] introduced an iterative Semantic Feedback module that refines prototypes based on prediction residuals, reducing semantic drift and improving GZSL harmonic mean on AWA2 and FLO. Yang et al. [29] combined semantic fusion with contrastive generation, synthesizing visual features for unseen classes and applying contrastive objectives to improve cluster compactness and reduce hubness, achieving state-of-the-art results across multiple benchmarks. Chen and Yeh [30] proposed indirect visual-semantic alignment that uses semantic information during training but performs inference entirely in the visual feature space, avoiding brittle prototype coupling at test time.

These methods collectively show that alignment quality—not model capacity alone—is the primary ZSL performance determinant, a lesson directly applicable to automotive prompt design.

C. Contrastive Vision-Language Models

Contrastive vision-language pretraining created a paradigm shift. CLIP [31] trains dual image and text encoders via InfoNCE loss over 400 million web-crawled image-text pairs; zero-shot recognition reduces to cosine similarity between image features and natural language prompt embeddings. LiT [32] improves efficiency by freezing a pretrained image encoder and tuning only the text head, preserving cross-modal alignment at lower compute cost. SigLIP [33] replaced softmax contrastive loss with a sigmoid binary objective, treating each image-text pair independently and eliminating the negative-pair competition that suppresses dense alignment; SigLIP2 [34] extended this to multilingual pretraining with improved localization and semantic compositionality. Multimodal architectures PaLI-3 [35] combined SigLIP encoders with autoregressive language decoder backbones, enabling unified grounding, captioning, and dense prediction. CoOp [36] further automated prompt engineering in CLIP-style models through learnable context token sequences optimized end-to-end, delivering significant gains over hand-crafted prompts in few-shot and cross-domain adaptation without modifying backbone weights.

IV. APPLYING ZSL TO VEHICLE RECOGNITION

A. From Closed-Set to Open-World VMMR

The progression from closed-set to open-world vehicle recognition traces a natural arc. Classical pipelines and early

CNN systems assumed a fixed label space at test time. The BoxCars verification protocol [8] approximates an open-set scenario through pairwise similarity decisions between instances, including rare near-unseen classes, but does not provide semantic labels for new classes. Vazquez-Santiago et al. [37] explicitly formalized VMMR as an open-set problem and introduced new-class discovery evaluation, decoupling the ability to reject unknown inputs from the ability to name them. ZSL provides the missing ingredient: semantic side information that enables a model to name a class it has never seen in images, completing the transition to open-world VMMR.

B. Existing Zero-Shot Work in the Vehicle Domain

Direct ZSL application to VMMR remains sparse. The GPT-4V evaluation [16] established the first empirical zero-shot baseline, confirming coarse-level brand identification capability without task-specific training but revealing insufficient reliability for deployment. In adjacent vehicle analysis, Luo et al. [38] developed FCLIP-VC, integrating Fourier-domain feature modulation into CLIP for zero-shot vehicle counting and localization, demonstrating that domain-specific architectural adaptations to CLIP improve transfer to vehicle-domain tasks. In the military recognition domain, hybrid pipelines combining object detectors with CLIP text-image alignment have been applied to classify unseen categories such as armored carriers using domain-specific prompts, outperforming supervised baselines in open-world generalization [38]. These results collectively confirm the viability of the ZSL approach for vehicles while underscoring the absence of a rigorous, standardized zero-shot benchmark for civilian VMMR.

TABLE II. Representative VMMR and ZSL methods by family, backbone, dataset, accuracy, and open-world readiness

Category	Sub-category	Method	Key Innovation	Highlights Dataset	Acc. (%)	Open-World readiness
Traditional Features	Hand-crafted descriptors	BoE [6]	Spatial co-occurrence BoW; multi-keypoint (KAZE, SURF, FAST, BRISK) + HOG; multiclass SVM	NTOU-MMR	SoTA (pre-DL)	None
		HOG + GIST [7]	1,664-D descriptor (1,152 HOG + 512 GIST over 4 scales); one-vs-one SVM; 13.9 fps on CPU	NTOU-MMR	97.89	None
CNN-Based Recognition	Geometry-aware normalization	BoxCars [8]	3D bounding box auto-calibration from fixed camera; surface patch unpacking; viewpoint vector injection	BoxCars	83.2	Partial (verify)
	Hierarchical coarse-to-fine	2B-2S [9]	Make-branch (DenseNet-201) + model-branch decoupling; Gain score quantifies inter-make confusion reduction	DVMM	93.95	None
	Part-aware agents	DeepCar 5.0 [3]	4 CNN agents (headlamp, grille, bumper, fog); blackboard consensus voting; occlusion / weather robust	DeepCar 5.0	92.14	None
		Lu et al. [10]	Dilated convolution part grouping; cross-channel feature optimization; ~140 fps on GPU	Stanford Cars	94.5	None
	Scale / taxonomy	Nafzi et al. [11]	Web-crawl multi-year multi-view images; hierarchical clustering infers 4,730-class taxonomy	Custom	71.1*	None
Vision Transformer	Supervised fine-tuning	Taki & Zemmouri [13]	ViT-Base/16 fine-tuned from ImageNet-21k; global self-attention over 16x16 patches; outperforms VGG16	Vehicle type	99.3	Closed
	Non-visual / radar	Liu et al. [14]	Carrier-free UWB radar echoes converted to 2D images; ViT outperforms CNN across all SNR levels	UWB dataset	—	Closed
	Self-supervised	DINOv2 [15]	Patch-level self-supervision without labels; dense features; strong zero-shot retrieval across domains	Multi-domain	—	Partial (backbone)
Vision-Language Models	Zero-shot evaluation	GPT-4V [16]	First ZS VMMR baseline on CompCars; no fine-tuning; brand bias + multi-vehicle failure documented	CompCars	Baseline	High (biased)
	Contrastive pretraining	CLIP [31]	InfoNCE softmax loss; 400M image-text pairs; open-vocab ZS recognition via cosine similarity	Web-scale	—	High
		SigLIP2 [34]	Sigmoid binary loss; multilingual pretraining; improved localization and dense semantic features	Web-scale	—	High
	Prompt adaptation	CoOp [36]	Learnable context tokens optimized end-to-end; significant gains over hand-crafted prompts; no backbone change	Multiple	—	High
ZSL / GZSL Methods	Semantic embedding	SSE [25]	Unseen class = convex combination of seen embeddings; structural similarity transfer	AWA, CUB	—	High
		Deep Emb. [26]	Max-margin compatibility function; deep visual-semantic space alignment	CUB, AWA	—	High
	Semantic refinement	Kim et al. [27]	Attribute-relevant feature extraction; irrelevant visual information suppressed during training	CUB, SUN	—	High
		Yang et al. [29]	Contrastive generation of unseen-class visual features; cluster compactness; reduces hubness	Multiple	—	High
	Indirect alignment	Chen & Yeh [30]	Semantic info used in training only; inference in visual feature space; avoids test-time prototype coupling	Multiple	—	High

V. GAPS AND FUTURE RESEARCH DIRECTIONS

Cross-examining both literatures reveals five interrelated gaps. Each is presented as: gap → why it matters → candidate research direction.

G1 – No GZSL benchmark for VMMR. *Why it matters:* All reviewed VMMR datasets use closed-set splits; no vehicle benchmark defines seen/unseen class partitions with semantic descriptors or reports harmonic mean H, making it impossible to track progress on zero-shot generalization or compare ZSL methods. *Direction:* Adapt large-scale benchmarks (VMMRDB, 495 classes; DVMM, 326 models) by defining held-out unseen splits annotated with automotive attribute descriptors—body type, headlamp style, grille topology, era—and adopting GZSL harmonic mean as the primary metric [4], [5].

G2 – Lack of domain-specific automotive prompts. *Why it matters:* The GPT-4V baseline [16] confirms that generic prompts such as “a photo of a [make]” conflate inter-make look-alikes due to training corpus brand bias. Fine-grained separability requires prompts encoding discriminative part-level cues: daytime running light (DRL) signature shape, grille mesh pattern, hood crease geometry, and era indicators. *Direction:* Compare manual attribute prompts, CoOp-learned context tokens [36] (which automate prompt engineering in models like CLIP by learning optimal context tokens), and LLM-generated descriptions; explore attribute-centric prompt fusion through cross-attention in the text encoder to compose symbolic discriminative descriptors between similar models.

G3 – Flat scoring amplifies inter-make confusion. *Why it matters:* Open-vocabulary scoring over hundreds of model classes forces all candidates to compete in a single pool, amplifying inter-make confusions because brands with similar design languages yield overlapping visual cues and prompt similarities. The 2B-2S CNN [9] shows that coarse-to-fine decomposition reduces inter-make confusion in supervised settings; an analogous mechanism is largely missing for ZSL-VMMR. *Direction:* Design a hierarchical ZSL head that first ranks makes using brand-level prompts and a DINOv2-pretrained visual encoder with a supervised linear probe (make-level routing), then resolves model identity within the top-K makes using SigLIP2 [34] similarity scoring; Fig. 5 operationalizes this make-model routing with track-level aggregation and calibration/unknown rejection to limit error propagation under surveillance shift. To further reduce propagation, include calibrated thresholds and an explicit “unknown” option at the make stage; optionally apply region-aware similarity on discriminative ROIs and aggregate across frames for surveillance robustness.

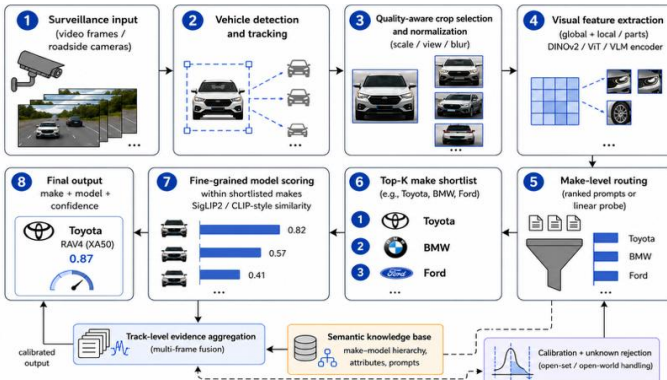


Fig. 5. Conceptual architecture for zero-shot VMMR

G4 – Domain shift degrades semantic alignment. *Why it matters:* ZSL benchmarks use curated web images; operational VMMR involves low-resolution, motion-blurred, night-time, and partially occluded surveillance footage [3]. The domain gap between VLM pretraining corpora and ITS imagery degrades both visual feature quality and prompt-image alignment. *Direction:* Study augmentation pipelines targeting surveillance degradations, viewpoint normalization inherited from geometry-aware approaches [8], and lightweight test-time calibration for the semantic scoring layer; measure their interaction with zero-shot generalization specifically under the web-to-surveillance domain shift.

G5 – Untested accuracy-latency trade-off on edge hardware. *Why it matters:* ITS pipelines operate on resource-constrained hardware where the inference cost of large VLMs is prohibitive. Frozen encoders (SigLIP2 [34], DINOv2 [15]) and parameter-efficient adaptation methods (LoRA [39]) reduce retraining cost, but their accuracy-latency profiles on edge devices such as NVIDIA Orin Nano have not been characterized for VMMR. *Direction:* Benchmark zero-shot VMMR pipelines end-to-end on representative edge hardware, identifying the minimum viable encoder size for real-time or near-real-time operation while satisfying target accuracy thresholds.

G5 – Untested accuracy-latency trade-off on edge hardware. *Why it matters:* ITS pipelines operate on resource-constrained platforms where the inference cost of large vision encoders can be prohibitive. SigLIP2 [34] provides explicit size variants—ViT-B 86M, ViT-L 303M, So400m 400M, and g 1B parameters—enabling systematic scaling studies. DINOv2 [15] similarly uses a ViT-g backbone with ~1.1B parameters and distills to smaller models, motivating profiling of distilled variants under deployment constraints. Although frozen encoders and LoRA [39] reduce retraining cost, their accuracy-latency profiles for VMMR are not characterized on edge devices such as Jetson Orin Nano. *Direction:* Benchmark zero-shot VMMR pipelines end-to-end on representative edge hardware, reporting params, FLOPs (at fixed resolution), peak memory, and measured latency/FPS under standardized FP16/INT8 settings, to identify the minimum viable encoder scale for real-time or near-real-time operation

VI. CONCLUSION

This paper provides a focused survey at the intersection of Vehicle Make and Model Recognition and Zero-Shot Learning. Tracing VMMR from hand-crafted Bag-of-Expressions pipelines through part-aware CNN architectures to transformer and vision-language approaches reveals a consistent pattern: closed-set accuracy on benchmark splits improves steadily, but structural generalization to unseen vehicle classes remains unaddressed. The ZSL literature, evolving from attribute-embedding methods through semantic refinement strategies to large-scale contrastive pretraining frameworks, now offers the semantic grounding machinery required to close this gap.

Our cross-analysis yielded three structured contributions: a taxonomy of VMMR method families organized by open-world readiness (Table II), a characterization of evaluation protocols from closed-set accuracy to GZSL harmonic mean, and five actionable research gaps paired with candidate ZSL-derived directions. Addressing gaps G1–G5—through standardized GZSL benchmarks, automotive prompt banks,

hierarchical make-aware inference heads, surveillance-domain calibration, and edge efficiency benchmarks—constitutes a concrete and high-impact agenda that can produce VMMR systems simultaneously accurate on seen classes, generalizable to unseen ones, robust under operational conditions, and deployable on ITS edge hardware.

VII. REFERENCES

- [1] Azzedine Boukerche și X. Ma, „Vision-based Autonomous Vehicle Recognition: A New Challenge for Deep Learning-based Systems”, *ACM Comput. Surv.*, vol. 54, no. 4, pp. 1-37, May 2022, doi: 10.1145/3447866.
- [2] X. Ni și H. Huttunen, „Vehicle Attribute Recognition by Appearance: Computer Vision Methods for Vehicle Type, Make and Model Classification”, *J. Signal Process. Syst.*, vol. 93, no. 4, pp. 357-368, April 2021, doi: 10.1007/s11265-020-01567-6.
- [3] A. Amirkhani și A. H. Barshooi, „DeepCar 5.0: Vehicle Make and Model Recognition Under Challenging Conditions”, *IEEE Trans. Intell. Transp. Syst.*, vol. 24, no. 1, pp. 541-553, January 2023, doi: 10.1109/tits.2022.3212921.
- [4] F. Pourpanah *et al.*, „A Review of Generalized Zero-Shot Learning Methods”, *IEEE Trans. Pattern Anal. Mach. Intell.*, pp. 1-20, 2022, doi: 10.1109/tpami.2022.3191696.
- [5] S. Rahman, S. Khan, și F. Porikli, „A Unified Approach for Conventional Zero-Shot, Generalized Zero-Shot, and Few-Shot Learning”, *IEEE Trans. Image Process.*, vol. 27, no. 11, pp. 5652-5667, November 2018, doi: 10.1109/TIP.2018.2861573.
- [6] A. A. Jamil, F. Hussain, M. H. Yousaf, A. M. Butt, și S. A. Velastin, „Vehicle Make and Model Recognition using Bag of Expressions”, *Sensors*, vol. 20, no. 4, p. 1033, February 2020, doi: 10.3390/s20041033.
- [7] A. J. Siddiqui, A. Mammeri, și A. Boukerche, „Real-Time Vehicle Make and Model Recognition Based on a Bag of SURF Features”, *IEEE Trans. Intell. Transp. Syst.*, vol. 17, no. 11, pp. 3205-3219, November 2016, doi: 10.1109/tits.2016.2545640.
- [8] J. Sochor, A. Herout, și J. Havel, „BoxCars: 3D Boxes as CNN Input for Improved Fine-Grained Vehicle Recognition”, in *2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, Las Vegas, NV, USA: IEEE, June 2016, pp. 3006-3015. doi: 10.1109/CVPR.2016.328.
- [9] Y. Lyu, I. Schiopu, B. Cornelis, și A. Munteanu, „Framework for Vehicle Make and Model Recognition—A New Large-Scale Dataset and an Efficient Two-Branch-Two-Stage Deep Learning Architecture”, *Sensors*, vol. 22, no. 21, p. 8439, November 2022, doi: 10.3390/s22218439.
- [10] L. Lu, Y. Cai, H. Huang, și P. Wang, „An efficient fine-grained vehicle recognition method based on part-level feature optimization”, *Neurocomputing*, vol. 536, pp. 40-49, June 2023, doi: 10.1016/j.neucom.2023.03.035.
- [11] M. Nafzi, M. Brauckmann, și T. Glasmachers, „Data Augmentation and Clustering for Vehicle Make/Model Classification”, vol. 1228, 2020, pp. 334-346. doi: 10.1007/978-3-030-52249-0_24.
- [12] „An Image is Worth 16x16 Words- Transformers for Image Recognition at Scale”.
- [13] „Vehicle Image Classification Method Using Vision Transformer”, in *Lecture Notes in Networks and Systems*, Cham: Springer Nature Switzerland, 2023, pp. 221-230. doi: 10.1007/978-3-031-43520-1_19.
- [14] X. Liu, Y. Zhu, S. Zhang, S. Chen, X. Li, și L. Zhu, „Vehicle Recognition Based on the Carrier-Free UWBR Using ViT Model and 2D-Echo Feature Images”, in *2023 5th International Conference on Computer Communication and the Internet (ICCCI)*, Fujisawa, Japan: IEEE, June 2023, pp. 42-46. doi: 10.1109/ICCCI59363.2023.10210164.
- [15] M. Oquab *et al.*, „DINOv2: Learning Robust Visual Features without Supervision”, 2 February 2024, *arXiv*: arXiv:2304.07193. doi: 10.48550/arXiv.2304.07193.
- [16] C. Kerdvibulvech, „Multimodal AI model for zero-shot vehicle brand identification”, *Multimed. Tools Appl.*, vol. 84, no. 27, pp. 33125-33144, January 2025, doi: 10.1007/s11042-024-20559-3.
- [17] J. Krause, J. Deng, M. Stark, și L. Fei-Fei, „Collecting a Large-Scale Dataset of Fine-Grained Cars”.
- [18] L. Yang, P. Luo, C. C. Loy, și X. Tang, „A Large-Scale Car Dataset for Fine-Grained Categorization and Verification”, 24 September 2015, *arXiv*: arXiv:1506.08959. doi: 10.48550/arXiv.1506.08959.
- [19] H. Liu, Y. Tian, Y. Wang, L. Pang, și T. Huang, „Deep Relative Distance Learning: Tell the Difference between Similar Vehicles”, in *2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, Las Vegas, NV: IEEE, June 2016, pp. 2167-2175. doi: 10.1109/CVPR.2016.238.
- [20] X. Liu, W. Liu, T. Mei, și H. Ma, „A Deep Learning-Based Approach to Progressive Vehicle Re-identification for Urban Surveillance”, in *Computer Vision – ECCV 2016*, vol. 9906, B. Leibe, J. Matas, N. Sebe, și M. Welling, Ed., in *Lecture Notes in Computer Science*, vol. 9906, Cham: Springer International Publishing, 2016, pp. 869-884. doi: 10.1007/978-3-319-46475-6_53.
- [21] Y. Lou, Y. Bai, J. Liu, S. Wang, și L. Duan, „VERI-Wild: A Large Dataset and a New Method for Vehicle Re-Identification in the Wild”, in *2019 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, Long Beach, CA, USA: IEEE, June 2019, pp. 3230-3238. doi: 10.1109/CVPR.2019.00335.
- [22] Z. Tang *et al.*, „CityFlow: A City-Scale Benchmark for Multi-Target Multi-Camera Vehicle Tracking and Re-Identification”, 5 April 2019, *arXiv*: arXiv:1903.09254. doi: 10.48550/arXiv.1903.09254.
- [23] F. Tafazzoli, H. Frigui, și K. Nishiyama, „A Large and Diverse Dataset for Improved Vehicle Make and Model Recognition”, in *2017 IEEE Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, Honolulu, HI, USA: IEEE, July 2017, pp. 874-881. doi: 10.1109/CVPRW.2017.121.
- [24] M. A. Manzoor, Y. Morgan, și A. Bais, „Real-Time Vehicle Make and Model Recognition System”, *Mach. Learn. Knowl. Extr.*, vol. 1, no. 2, pp. 611-629, April 2019, doi: 10.3390/make1020036.
- [25] Z. Zhang și V. Saligrama, „Zero-Shot Learning via Semantic Similarity Embedding”, in *2015 IEEE International Conference on Computer Vision (ICCV)*, Santiago, Chile: IEEE, December 2015, pp. 4166-4174. doi: 10.1109/ICCV.2015.474.
- [26] L. Zhang, T. Xiang, și S. Gong, „Learning a Deep Embedding Model for Zero-Shot Learning”, in *2017 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, Honolulu, HI: IEEE, July 2017. doi: 10.1109/cvpr.2017.321.
- [27] J. Kim, K. Shim, și B. Shim, „Semantic Feature Extraction for Generalized Zero-Shot Learning”, *Proc. AAAI Conf. Artif. Intell.*, vol. 36, no. 1, pp. 1166-1173, June 2022, doi: 10.1609/aaai.v36i1.20002.
- [28] L. Zhang și G. Zhang, „Semantic Feedback for Generalized Zero-Shot Learning”, in *2024 4th International Conference on Neural Networks, Information and Communication (NNICE)*, Guangzhou, China: IEEE, January 2024, pp. 298-302. doi: 10.1109/NNICE61279.2024.10499130.
- [29] G. Yang, W. Sun, X. Liu, Y. Liu, și C. Wang, „Semantic Fusion and Contrastive Generation for Generalized Zero-Shot Learning”, *Int. J. Multimed. Inf. Retr.*, vol. 14, no. 3, sep. 2025, doi: 10.1007/s13735-025-00372-w.
- [30] Y.-H. Chen și M.-C. Yeh, „Indirect visual-semantic alignment for generalized zero-shot recognition”, *Multimed. Syst.*, vol. 30, no. 2, April 2024, doi: 10.1007/s00530-024-01313-z.
- [31] A. Radford *et al.*, „Learning Transferable Visual Models From Natural Language Supervision”, 26 February 2021, *arXiv*: arXiv:2103.00020. doi: 10.48550/arXiv.2103.00020.
- [32] „LiT - Zero-Shot Transfer with Locked-image text Tuning”.
- [33] X. Zhai, B. Mustafa, A. Kolesnikov, și L. Beyer, „Sigmoid Loss for Language Image Pre-Training”, in *2023 IEEE/CVF International Conference on Computer Vision (ICCV)*, Paris, France: IEEE, October 2023. doi: 10.1109/iccv51070.2023.01100.
- [34] M. Tschannen *et al.*, „SigLIP 2: Multilingual Vision-Language Encoders with Improved Semantic Understanding, Localization, and Dense Features”, 20 February 2025, *arXiv*: arXiv:2502.14786. doi: 10.48550/arXiv.2502.14786.
- [35] X. Chen *et al.*, „PaLI-3 Vision Language Models: Smaller, Faster, Stronger”, 17 October 2023, *arXiv*: arXiv:2310.09199. doi: 10.48550/arXiv.2310.09199.
- [36] K. Zhou, J. Yang, C. C. Loy, și Z. Liu, „Learning to Prompt for Vision-Language Models”, *Int. J. Comput. Vis.*, vol. 130, no. 9, pp. 2337-2348, sep. 2022, doi: 10.1007/s11263-022-01653-1.
- [37] D.-I. Vázquez-Santiago, H.-G. Acosta-Mesa, și E. Mezura-Montes, „Vehicle Make and Model Recognition as an Open-Set Recognition Problem and New Class Discovery”, *Math. Comput. Appl.*, vol. 28, no. 4, p. 80, July 2023, doi: 10.3390/mca28040080.
- [38] Y. Luo, Q. Zhang, J. Luo, S. Huang, X. Wan, și X. Nie, „Fourier-modulated CLIP for zero-shot vehicle counting”, *Sci. Rep.*, vol. 15, no. 1, p. 19732, June 2025, doi: 10.1038/s41598-025-04876-7.
- [39] E. J. Hu *et al.*, „LoRA: Low-Rank Adaptation of Large Language Models”, 16 October 2021, *arXiv*: arXiv:2106.09685. doi: 10.48550/arXiv.2106.09685.